DAY 1 · AWS LINUX LAB

Cron & Logrotate

Scheduled jobs & log rotation

Built a scheduled disk-usage report (`disk-report.sh`) via cron, then set up logrotate so its log never grows unbounded.

Along the way, migrated the whole pipeline to a dedicated `svc-disk-report` service account — verified end-to-end with a real forced rotation.

Local — your machine · Remote — the server · Config file · Concept · Gotcha · Best practice · Paths

What we covered

- ✓ Wrote `disk-report.sh` — timestamp + `df -h` output
- ✓ Scheduled it with `crontab -e` — 5-field time syntax, per-user schedules
- ✓ Why unbounded logs are a real problem, and how logrotate solves it
- ✓ Built a `/etc/logrotate.d/` config: daily rotation, compression, 7-day retention
- ✓ Debugged real logrotate errors (inline comments, insecure directory permissions)
- ✓ Migrated the entire pipeline — script, log folder, cron entry, logrotate `su` — to run as `svc-disk-report`
- ✓ Verified with a forced rotation: `disk-report.log.1.gz` was created

Full command sequence

Remote The script

```
sudo nano /usr/local/bin/disk-report.sh   # create the script as root
```

```
/usr/local/bin/disk-report.sh
#!/bin/bash
echo "Disk report at $(date)" >> /var/log/disk-report/disk-report.log   # timestamp header
df -h >> /var/log/disk-report/disk-report.log                           # -h: human-readable sizes
echo "---" >> /var/log/disk-report/disk-report.log                      # separator line
```

```
sudo chmod +x /usr/local/bin/disk-report.sh   # +x: make it executable
```

Concept

Cron's 5-field syntax

```
*  *  *  *  *  command
|  |  |  |  |
|  |  |  |  └─ day of week   (0-7, 0 and 7 = Sunday)
|  |  |  └──── month         (1-12)
|  |  └─────── day of month  (1-31)
|  └────────── hour          (0-23)
└───────────── minute        (0-59)
```

Remote

Schedule it

```
crontab -e   # -e: edit your own crontab
```

crontab entry

```
0 * * * * /usr/local/bin/disk-report.sh                    minute 0 of every hour = hourly, on the hour
```

```
crontab -l   # -l: list what's scheduled
```

Remote

Logrotate config

```
sudo nano /etc/logrotate.d/disk-report   # create the rotation rule
```

/etc/logrotate.d/disk-report

```
/var/log/disk-report/disk-report.log {          which log this rule covers
    su svc-disk-report svc-disk-report          rotate as this user + group
    daily                                       rotate once a day
    rotate 7                                    keep 7 old copies
    compress                                    gzip old copies
    missingok                                   no error if the log is missing
    notifempty                                  skip rotation if it's empty
    copytruncate                                copy, then empty the live file
}
```

Bugs hit & fixed

Bug · inline comments broke the config

CAUSE Unlike bash, logrotate doesn't treat `#` as "ignore the rest of the line" — `rotate 7 # comment` was read as one invalid value.

FIX Comments go on their own line, never after a setting.

Bug · insecure directory permissions

CAUSE logrotate (running as root) refused to touch a log whose parent directory wasn't strictly root-owned.

FIX Added the `su <user> <group>` directive so logrotate rotates as the account that owns the log.

Bug · file owned, but not the directory

CAUSE Rotation creates a new file (`disk-report.log.1`) in the parent directory — owning only the log file wasn't enough.

FIX Gave the log its own dedicated subfolder, owned entirely by the service account.

Remote Migrate to svc-disk-report

```
cd /var/log                                                    # work in /var/log
sudo mkdir -p disk-report                                      # -p: no error if it exists
sudo chown svc-disk-report:svc-disk-report disk-report         # hand folder to the svc
sudo mv disk-report.log disk-report/disk-report.log            # move log into its folder
sudo chown svc-disk-report:svc-disk-report disk-report/disk-report.log   # hand log to the svc
sudo crontab -u svc-disk-report -e                             # -u: the svc's crontab
```

Remote Test the full pipeline

```
sudo logrotate -d /etc/logrotate.d/disk-report   # -d: dry run, checks syntax only
sudo logrotate -f /etc/logrotate.d/disk-report   # -f: force a real rotation now
ls -la /var/log/disk-report/                     # -la: list all, long format; look for .log.1.gz
```

Where things live

What	Path
Report script	`/usr/local/bin/disk-report.sh`
Log directory (own folder per service)	`/var/log/disk-report/`
Logrotate config	`/etc/logrotate.d/disk-report`
Personal crontab	`crontab -e` / `crontab -l`
Another user's crontab	`sudo crontab -u <user> -e`

Best practices

Plan rotation up front
Any log that grows unattended needs a rotation policy — decide it before it's a problem.

One log folder per service
Matches real-world convention (`nginx` , `postgres` and others).

Check comment rules per tool
Config files aren't always inline-comment-safe like bash — read each tool's syntax rules.

Dry-run first
Always `logrotate -d` before forcing a real rotation.

Jobs run as their service
Background jobs belong to their dedicated service account, not your personal sudo account.

Use full paths in cron
Cron runs with a minimal environment, so absolute paths avoid “command not found”.

Part of the AWS/Linux sysadmin lab · see day1-naming-conventions for the full naming reference